

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

001 This supplement records the material that supports
002 the main paper without being required to follow its
003 argument: the full metric specification, the complete
004 validation matrix, the evaluation-set composition, per-
005 horizon results beyond the two principal horizons, and
006 the exploratory image-conditioned analysis. Section
007 numbering is independent of the main paper; cross-
008 references in the main text point here by name.

009 1. Region Partition

010 Derivation. For every generated sequence we average
011 optical-flow magnitude over the leading 12% of sam-
012 pled frame pairs and threshold it by Otsu’s method
013 into a dynamic region and its complement. The static
014 region is then reduced by a fixed morphological ero-
015 sion, and a 4% frame border is discarded; together
016 these form the ignored transition band, so boundary
017 pixels—where the partition is least certain and flow
018 least reliable—contribute to neither region. The same
019 procedure is applied to both tracks.

020 Why the estimate is restricted to an early window.
021 Deriving a partition from a sequence’s entire motion
022 field would be circular, since accumulated background
023 drift would itself be read as evidence that the affected
024 region is dynamic. Restricting the estimate to the early
025 window breaks that circularity in the direction that
026 matters: at the 60s horizon the partition is fixed from
027 the first 7.1s, before long-horizon drift has accumu-
028 lated, and the metrics are then evaluated over the re-
029 mainder.

030 Limitations. The partition is obtained automatically
031 rather than by annotation, so no inter-annotator agree-
032 ment is reported, and for the image-conditioned track
033 it is derived per generated sequence rather than once
034 from the shared source image. The early window is
035 a weaker guarantee than a first-frame annotation: a
036 system that drifts within the first 12% of its rollout
037 can influence its own partition. We therefore treat
038 partitions as measured quantities and report the con-
039 sequence directly—every measurement is recomputed
040 under systematic erosion and dilation of the boundary,
041 with the resulting response given in Table 1. A human-
042 verified, first-frame annotation protocol is the natural
043 strengthening of this component.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

2. Complete Validation Matrix

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s own parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of arctan-free angle $\theta = d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame centre. Expressing severity this way is what makes the geometric families comparable; a parameter expressed in degrees is not commensurate with one expressed in pixels, and treating them as though they were under-states angular corruptions by more than an order of magnitude at this resolution.

3. Whole-Frame and SNF-Bench Rank Correlation

4. Complementary Perceptual Checks

Neither check is aggregated into a factor, and neither is a preference study. Both answer one question: does an independent judge, with no access to our factors, order the same clip pairs the same way?

Pair selection. Pairs are chosen by rule rather than by eye. For each axis, candidate pairs of systems on

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth	-0.96	-0.82	-0.61	-0.64
VB-flick	-0.96	-0.82	-0.61	-0.64

Table 2. Rank disagreement at 60s. Spearman correlation between generic whole-frame metrics and SNF-Bench factors over method-level means. A positive correlation between apparent motion and background flow is the effect the benchmark isolates.

067 a shared prompt are binned by the relative gap in the
 068 relevant factor and sampled across three bands (0.05–
 069 0.20, 0.20–0.50, 0.50–1.00), so agreement is measured
 070 over the whole operating range rather than only where
 071 any metric would look good. Thirty pairs per axis are
 072 drawn.

073 Human study. Raters answer exactly two forced-
 074 choice questions—which video’s background moves
 075 more, and in which video the moving content slows or
 076 stops more—and are never asked which video is better,
 077 since “better” conflates the two failures the benchmark
 078 separates. Trial order and left/right presentation are
 079 randomised per rater, choices are resolved back to sys-
 080 tem identities server-side, and system names are never
 081 shown. Ties are permitted: forcing a choice on an in-
 082 distinguishable pair manufactures noise. Sixty trials
 083 per rater, three raters.

084 Vision–language judge. The same pairs, with the
 085 same wording, are put to an open-weight vision–
 086 language model [1]. Each pair is asked twice with the
 087 clips swapped; a judge answering by position rather
 088 than by content disagrees with itself, so agreement is
 089 computed only over self-consistent trials and the self-
 090 consistency rate is reported beside it. A judge that
 091 contradicts itself on a pair has expressed no opinion
 092 about that pair.

093 Reporting. Per axis we report the agreement rate
 094 with a binomial interval, the tie rate, and agreement
 095 resolved by metric gap. The last matters: a factor that
 096 agrees with observers only on large gaps is a weaker re-
 097 sult than one that agrees throughout, and pooling hides
 098 the difference.

category	definition
river_stream	channel water: rivers, streams, creeks, canals, w
ocean_waves	open water with wave action: surf, tides, co
precipitation	falling water or ice: rain, snowfall, and the
fire_smoke	combustion plumes: wildfire, campfire, hearth,
lava_volcanic	volcanic flow and ejecta: lava channels, ash co
windborne	air-carried particulates and wind-driven vegetation: dust, sand, blo
track/duration	n
i2v/5s	10
i2v/60s	30
i2v/120s	20
i2v/240s	5
t2v/5s	6
t2v/60s	23
t2v/120s	6
t2v/240s	4
track	source
i2v	meta
t2v	keyword
t2v	keyword-body
t2v	override
track	duration
t2v	60s

Table 3. Scene-category balance of the evaluation set, by track and horizon.

Method	status	setting	5s	
CausVid	public	native	V6 B	V23
Self-Forcing	public	native	V6 B	V23
Infinite-Forcing	public	native	V6 B	V23
Rolling-Forcing	public	native	V6 B	V24
Reward-Forcing	public	native	V12 B	V23
LongLive	public	native	V6 B	V23
Causal-Forcing	public	native	V6 B	V23
Method	status	setting	5s	
Causal-Forcing++ (2-step)	public	matched	V10 T10 X B	V30
Causal-Forcing++ (1-step)	public	matched	V10 T10 X B	V30 T
Causal-Forcing (framework)	public	matched	V10 T10 X B	V30
Causal-Forcing++ (2-step, native nfpb=1)	public	native	V10 T10 X B	V30
Self-Forcing	public	matched	V10 T10 X B	V30
CausVid	public	matched	V10 T10 X B	V30 T
Wan2.1-I2V-14B-480P	public	native	V10 T10 X B	
Wan2.2-I2V-A14B	public	native	V10 T10 X B	
LTX-Video 13B-0.9.8-distilled	public	native	V10 T10 X B	
Causal-Forcing++ (1-step, native)	public	native	.	V3

Table 4. Asset and score coverage for the audited public systems. T_n/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

5. Evaluation-Set Composition

099

6. Asset and Score Coverage

100

Coverage is counted from valid per-video measure-
 ments rather than from the presence of a results file,
 so an entry whose measurement run terminated early
 is reported as partial rather than complete.

101

102

103

104

7. Results at All Horizons

105

The main paper reports the two principal horizons.
 The 5s tier serves as an initialisation check and the
 240s tier is exploratory; both are given here for com-
 pleteness.

106

107

108

109

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 5. Complete T2V audit, all horizons. The main paper reproduces the 60s and 120s panels; the 5s tier is an initialisation check and the 240s tier is diagnostic. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

110 8. Signed Drift Attenuation

111 Drift attenuation is stored as a signed quantity and
 112 reported clipped to $[0, 1]$. Negative values are informa-
 113 tive rather than anomalous: where local flow opposes
 114 the estimated global field, compensation increases the
 115 measured magnitude. We report their incidence rather
 116 than suppressing it.

117 9. Deployment Sensitivity

118 10. Interpretation Changes

119 Cases are selected by a fixed rank-gap rule applied
 120 to every audited system, so no qualifying case can be
 121 omitted; systems exhibiting no qualifying gap simply
 122 do not appear.

11. Image-Conditioned Track

The image-conditioned analysis is exploratory and is
 not part of the main paper’s claims. Systems marked
 matched are evaluated under a common long-horizon
 wrapper rather than their released configuration, and
 their scores are read as deployment sensitivity rather
 than as the published method’s performance.

12. Additional Qualitative Material

References

[1] Qwen Team. Qwen3-VL technical report. arXiv
 preprint arXiv:2511.21631, 2025. 2

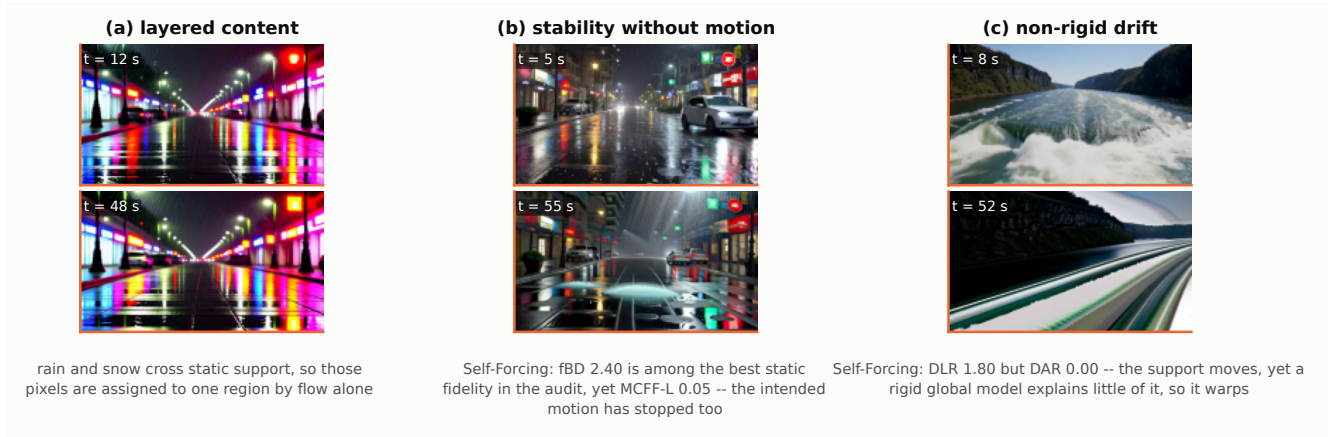


Figure 1. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.

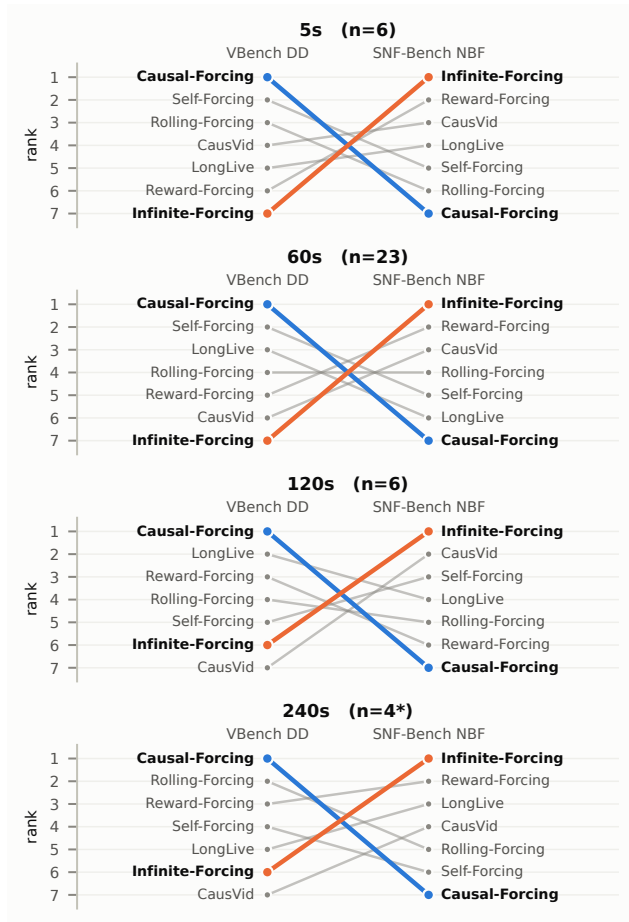


Figure 2. Rank disagreement at every audited horizon. Rank under a whole-frame motion score against rank under normalised background flow. Both quantities are independent of the global-motion estimator. Prompt counts are given per panel; the shorter and longer tiers are diagnostic.

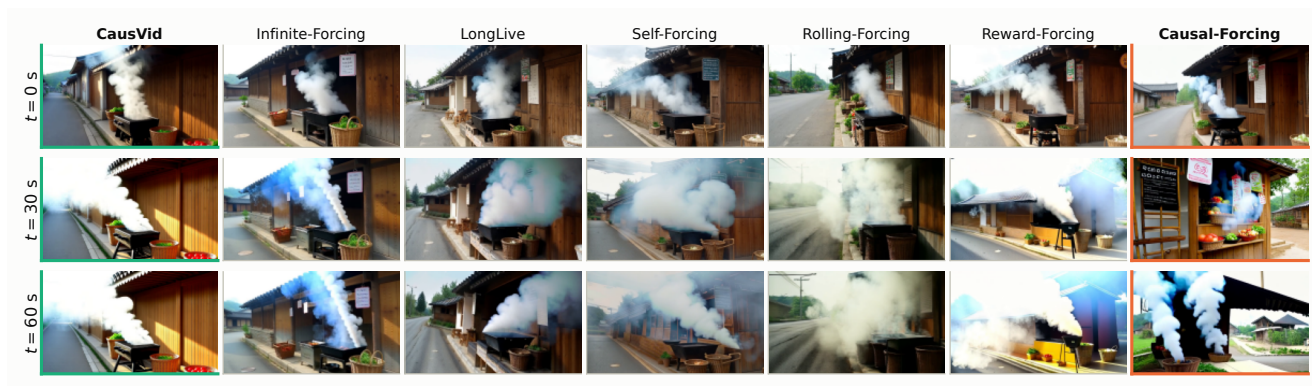


Figure 3. Every audited system on one prompt. Columns are ordered by increasing static-region drift. The two highlighted systems are the pair a whole-frame motion score cannot separate: the leftmost holds its support fixed while its smoke continues to rise, the rightmost has relocated the scene entirely by $t = 30$ s.

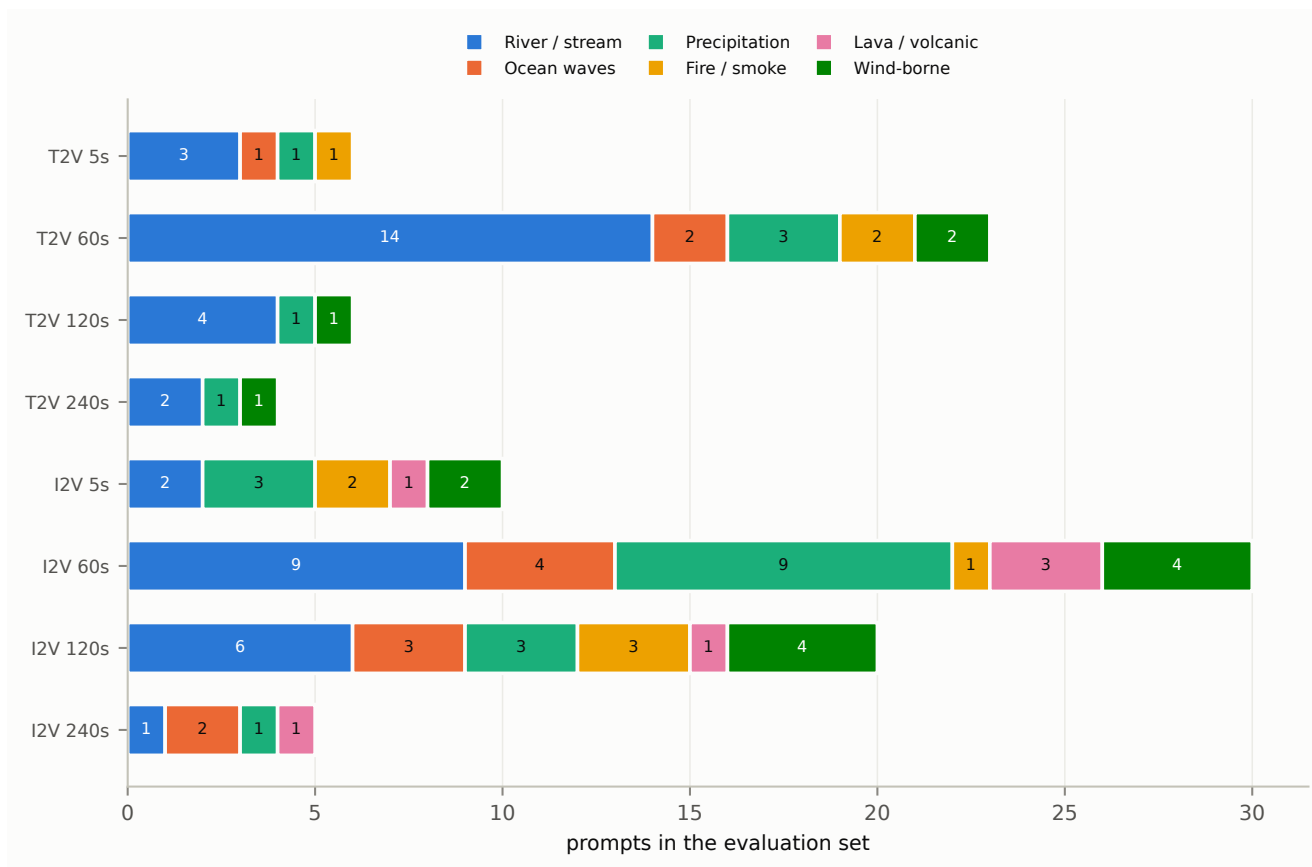


Figure 4. Composition of the evaluation set by scene category. Counts are prompts per category at each track and horizon.

Method	track	dur	n	negative	rate	most negative					
CausVid	i2v	120s	20	1	5.0%	-0.055					
CausVid	i2v	240s	5	0	0.0%	0.000					
CausVid	i2v	5s	10	6	60.0%	-0.450					
CausVid	i2v	60s	30	2	6.7%	-0.084					
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.012					
Causal-Forcing++ (1-step)	i2v	240s	5	0	0.0%	0.175					
Causal-Forcing++ (1-step)	i2v	5s	10	0	0.0%	0.041					
Causal-Forcing++ (1-step)	i2v	60s	30	0	0.0%	0.013					
Causal-Forcing (framework)	i2v	120s	20	3	15.0%	-0.171					
Causal-Forcing (framework)	i2v	240s	5	0	0.0%	0.046					
Causal-Forcing (framework)	i2v	5s	10	2	20.0%	-0.043					
Causal-Forcing (framework)	i2v	60s	30	0	0.0%	0.008					
Causal-Forcing++ (2-step)	i2v	120s	20	0	0.0%	0.008					
Causal-Forcing++ (2-step)	i2v	240s	5	0	0.0%	0.053					
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.102					
Causal-Forcing++ (2-step)	i2v	60s	30	0	0.0%	0.005					
Causal-Forcing++ (1-step, native)	i2v	120s	20	1	5.0%	-0.062					
Causal-Forcing++ (1-step, native)	i2v	60s	30	3	10.0%	-0.770					
Causal-Forcing++ (2-step, native nfpb=1)	i2v	120s	20	3	15.0%	-0.192					
Causal-Forcing++ (2-step, native nfpb=1)	i2v	240s	5	2	40.0%	-0.164					
Causal-Forcing++ (2-step, native nfpb=1)	i2v	5s	10	1	10.0%	-0.133					
Causal-Forcing++ (2-step, native nfpb=1)	i2v	60s	30	2	6.7%	-0.014					
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	4	40.0%	-0.031					
Self-Forcing	i2v	120s	20	3	15.0%	-0.131					
Self-Forcing	i2v	240s	5	2	40.0%	-0.012	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Self-Forcing	i2v	5s	10	2	20.0%	-0.036	5s	+2.47	-0.61	-0.18	-0.01
Self-Forcing	i2v	60s	30	8	26.7%	-0.096	60s	-0.76	-0.95	-0.15	-0.08
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.012	120s	-1.71	-8.46	-0.61	-0.17
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.012	240s	+4.06	+0.88	-0.06	-0.28
CausVid	t2v	120s	6	0	0.0%	-0.490	60s	-5.05	+1.80	+0.05	-0.20
CausVid	t2v	240s	4	1	25.0%	-0.384	120s	-6.34	+2.00	+0.41	-0.14
CausVid	t2v	5s	6	2	33.3%	-0.093					
CausVid	t2v	60s	23	0	0.0%	-0.175					
Causal-Forcing	t2v	120s	6	0	0.0%	0.031					
Causal-Forcing	t2v	240s	4	0	0.0%	0.158					
Causal-Forcing	t2v	5s	6	2	33.3%	-0.275					
Causal-Forcing	t2v	60s	23	3	13.0%	-0.317					
Infinite-Forcing	t2v	120s	6	0	0.0%	0.023					
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158					
Infinite-Forcing	t2v	5s	6	0	0.0%	0.031					
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359					
LongLive	t2v	120s	6	1	16.7%	-1.076					
LongLive	t2v	240s	4	0	0.0%	0.022					
LongLive	t2v	5s	6	1	16.7%	-0.139					
LongLive	t2v	60s	23	2	8.7%	-0.051					
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116					
Reward-Forcing	t2v	240s	4	0	0.0%	0.002					
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145					
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317					
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353					
Rolling-Forcing	t2v	240s	4	0	0.0%	0.043					
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068					
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038					
Self-Forcing	t2v	120s	6	2	33.3%	-0.130					
Self-Forcing	t2v	240s	4	1	25.0%	-0.153					
Self-Forcing	t2v	5s	6	1	16.7%	-0.046					
Self-Forcing	t2v	60s	23	4	17.4%	-1.117					

Table 6. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow.

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 8. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	–	0.67	0.86	0.67	0.15
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (framewise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, native nfpb=1)	native	9.94	6.91	–	0.81	0.98	0.75	0.11
Self-Forcing	matched	17.3	5.52	–	0.74	0.88	0.67	0.14
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, native)	native	10.4	5.25	–	0.52	0.97	0.83	0.14

Table 9. SNF-Bench audit, I2V track. Setting is stated per row: native systems run under their authors’ configuration, matched systems under the common long-horizon wrapper and are read as a deployment stress test rather than as the published method’s performance. n is stated per horizon and is common to every row of that panel.